

CareerGPS

System Analysis & Design Document

AI-Powered Career Navigation Platform

Document	System Analysis & Design
Version	1.0 - Initial Release
Program	Rwaad Masr Digital Scholarship
Team	12 Members (4 FE 2 FS 6 Agentic AI)
Architecture	Microservices + Agentic AI Layer
Date	April 2025

1. Problem Statement & Objectives

1.1 Problem Statement

Thousands of Egyptian and Arab graduates face a critical orientation failure: they know they need to learn, but not what, in what order, or from where - relative to their own unique background. Generic roadmaps ignore existing experience, learning resources are fragmented across hundreds of platforms, and no accountability mechanism exists to sustain momentum over months of self-directed learning.

CareerGPS solves this by combining AI-powered profile analysis, dynamic roadmap generation, curated resource matching, and a gamified learning calendar - all personalized to the individual.

1.2 Project Objectives

#	Objective	Description	Success Metric
O1	Personalized Roadmaps	Generate tailored roadmaps from user profiles using AI	Roadmap ready in <5 sec; satisfaction -4/5
O2	Eliminate Resource Noise	Curate -1 course per skill; compare playlists	User reports clarity improvement -80%
O3	Sustain Accountability	Calendar + motivational push notifications	Study adherence rate - 60% active users
O4	Job Readiness	Interview sim + ATS CV builder + skill exams	-70% report increased interview confidence
O5	On-time Delivery	Ship MVP Phase 1 within 4 months	All FR-01 to FR-11 live and tested

2. Use Case Diagram & Descriptions

2.1 System Actors

Actor	Type	Description
Job Seeker	Primary	Core user - uploads CV, generates roadmap, learns, takes exams
Career Explorer	Primary	Explores new career paths via simulation before committing
Admin	Secondary	Manages users, content, course catalog, system monitoring
Claude AI (Anthropic)	External System	AI engine for roadmaps, exams, CV building, interview feedback
GitHub API	External System	Repository data and inferred technical skills
LinkedIn (PDF Export)	External System	Work experience and skills from user-exported profile
Job Boards API	External System	Job listings matched to user skill set

2.2 Use Case Diagram

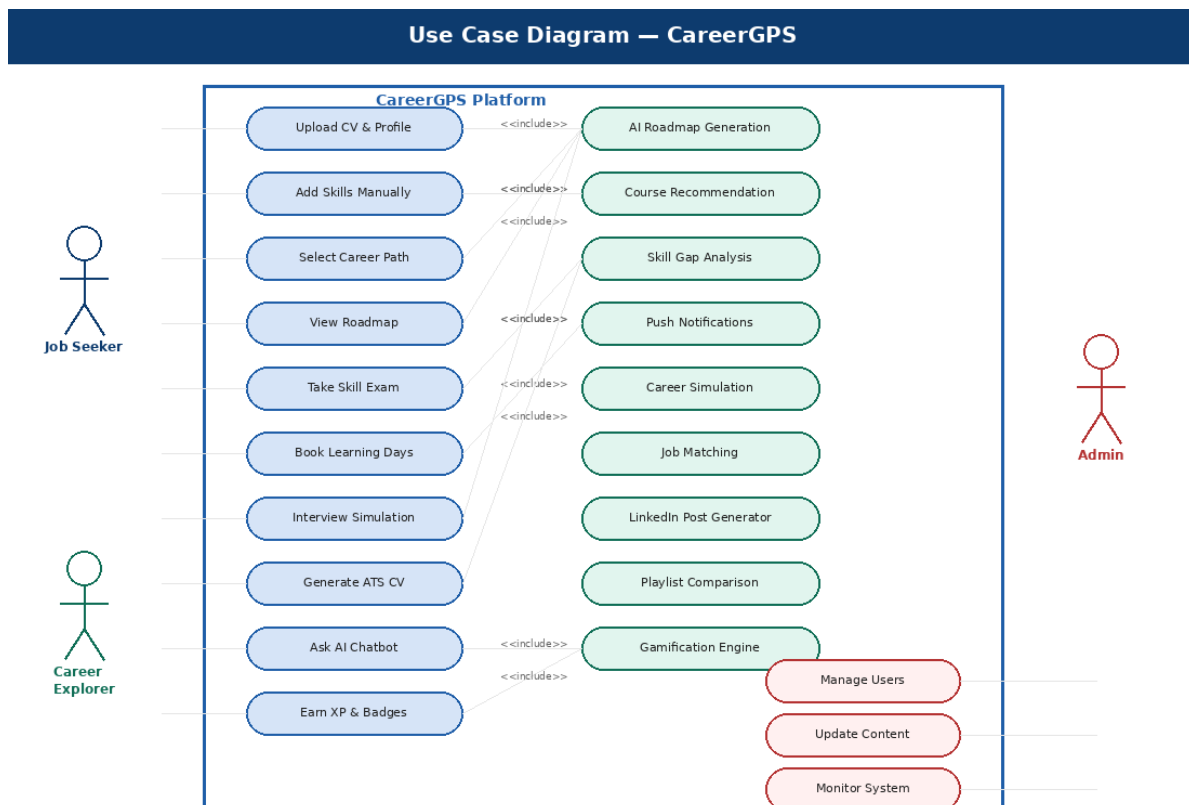


Figure 1 - CareerGPS System Use Case Diagram

2.3 Use Case Descriptions

UC-01: Upload & Parse CV

Use Case ID	UC-01
Actor	Job Seeker
Description	User uploads PDF CV; AI extracts skills, experience, education automatically.
Precondition	User authenticated
Postcondition	Profile populated with extracted data
Main Flow	1. User navigates to Profile page 2. User uploads CV (PDF -5MB) 3. System forwards to CV Parser AI agent 4. AI extracts skills, experience, education 5. User reviews and confirms extracted data

UC-02: Generate Personalized Roadmap

Use Case ID	UC-02
Actor	Job Seeker
Description	User selects career; AI generates tailored roadmap with ordered skill nodes and course links.
Precondition	Profile has skill data
Postcondition	Roadmap saved, courses linked per node
Main Flow	1. User selects target career path 2. System calls Roadmap Agent with profile + career 3. Claude AI returns structured roadmap JSON 4. System persists roadmap and fetches courses 5. Roadmap displayed with progress tracker

UC-03: Take Skill Exam

Use Case ID	UC-03
Actor	Job Seeker
Description	User takes AI-generated exam for a skill to assess knowledge and identify gaps.
Precondition	Skill exists in user roadmap
Postcondition	Exam results and gap report saved
Main Flow	1. User selects skill - 'Take Exam' 2. Exam Agent generates MCQ + open-ended questions 3. User submits answers 4. AI evaluates and scores answers 5. Gap analysis report and resource recommendations shown

UC-04: Interview Simulation

Use Case ID	UC-04
Actor	Job Seeker
Description	User simulates a real job interview with role-specific questions and receives AI feedback on each answer.
Precondition	User has a roadmap with target career
Postcondition	Session saved, overall feedback report generated
Main Flow	1. User starts Interview Simulation for career 2. Interview Agent generates role-specific questions 3. User answers each question within time limit 4. AI evaluates answer quality, relevance, clarity 5. Final report: score + strengths/weaknesses + improvement tips

UC-05: Generate ATS-Friendly CV

Use Case ID	UC-05
Actor	Job Seeker
Description	User pastes a job description; AI builds a tailored, keyword-optimized CV with ATS score.
Precondition	User has completed profile
Postcondition	CV PDF generated, stored, downloadable
Main Flow	1. User pastes or uploads job description 2. AI extracts keywords with importance weights 3. CV Builder Agent creates CV matching profile + keywords 4. ATS Scorer evaluates keyword match percentage 5. User previews and downloads CV as PDF

3. Functional & Non-Functional Requirements Summary

Full requirements are in the Requirements Specification Document (v1.0). Below is the module-level summary.

3.1 Functional Requirements by Module

Module	Core Capabilities
Profile	CV parsing (AI), LinkedIn PDF import, GitHub API integration, manual skill management
Roadmap	AI-generated personalized roadmap, phase ordering, progress tracking, regeneration on change
Courses	Per-skill course recommendation, YouTube playlist comparison, curated + AI-suggested sources
Calendar	Study schedule builder, reminder configuration, timezone support
Assessments	AI exam generation (MCQ + open-ended), gap analysis, targeted resource recommendations
CV Builder	ATS keyword extraction from JD, tailored CV generation, from-scratch mode, PDF export
Interview	Role-specific mock interview, AI answer evaluation, scoring, feedback report per answer
Gamification	XP system, badge library, level progression, leaderboard, streak tracker
Career Sim	Guided hands-on career simulation tasks (TryHackMe-style) across multiple careers
Notifications	FCM push + email; AI-generated motivational messages on scheduled study days
Jobs	Skill-based job matching from external APIs (Wuzzuf, Indeed, RemoteOK)
Admin	User management, content moderation, platform analytics, system health dashboard

3.2 Non-Functional Requirements

Category	Requirement
Performance	Roadmap generation <5 sec; CV parsing <10 sec; API response P95 <2 sec
Scalability	10,000 concurrent users; horizontal pod autoscaling; stateless services
Availability	99.5% uptime SLA; automated health-check restart; circuit breaker on AI calls
Security	AES-256 at rest; TLS 1.3 in transit; JWT + refresh rotation; OWASP Top 10 compliance
Usability	Core task (CV - roadmap) in ~3 steps; WCAG 2.1 AA; mobile-first responsive design
Localization	Full Arabic (RTL) + English (LTR) support; locale-aware date/number formatting
Maintainability	Microservices; independent deployment per service; 80%+ unit test coverage target
Privacy	GDPR-compliant; right to deletion; data export API; minimal PII in AI prompts

4. Software Architecture

4.1 Architecture Style: Microservices + Agentic AI

CareerGPS uses a 6-layer microservices architecture. Domain services are independently deployable and communicate via REST APIs synchronously and RabbitMQ message queues asynchronously for AI-heavy operations. The Agentic AI Layer orchestrates multi-step LLM workflows using LangChain/LangGraph with Claude as the inference engine.

- Independent scaling - AI services scale separately from the UI and CRUD services.
- Fault isolation - a failing Course Service does not cascade to the Roadmap Service.
- Team parallelism - Frontend, Full-Stack, and AI teams deploy independently on separate pipelines.
- Extensibility - new features require only a new microservice, not modification of existing ones.

4.2 Architecture Diagram

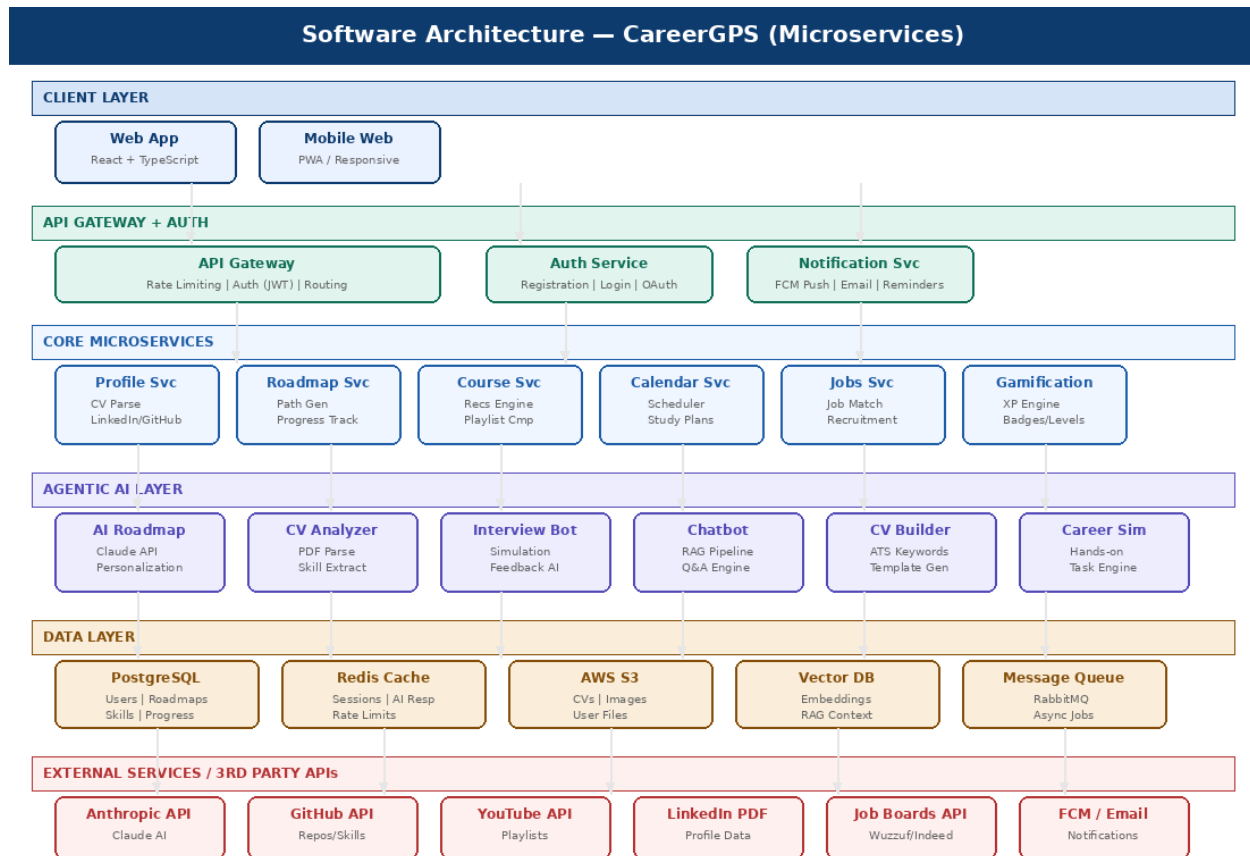


Figure 2 - CareerGPS 6-Layer Microservices Architecture

4.3 Component Descriptions

Component	Technology	Responsibility
Client Layer	React + TypeScript PWA	UI rendering, state management, REST/WebSocket calls to API Gateway
API Gateway	Node.js + Express	Request routing, JWT auth validation, rate limiting, load balancing
Auth Service	Clerk / custom JWT	Registration, login, OAuth, refresh token rotation
Profile Service	Node.js microservice	CV upload, AI parsing, GitHub integration, skill management
Roadmap Service	Node.js microservice	Roadmap generation orchestration, node tracking, progress updates
Course Service	Node.js microservice	Course recommendations, playlist comparison engine
AI Agent Layer	Python + LangChain/LangGraph	Orchestrates multi-step Claude AI workflows for all AI features
Gamification Svc	Node.js microservice	XP calculation, badge awards, streak tracking, leaderboard
Notification Svc	FCM + SendGrid	Push notifications, motivational messages, email delivery
PostgreSQL	Database	Primary relational store for all user and domain data
Redis	Cache / Queue	Session caching, AI response caching, rate limit counters
AWS S3 / R2	Object Storage	CV files, generated PDFs, profile images
Vector DB	Pinecone / Chroma	Text embeddings for RAG pipeline powering chatbot and recommendations

5. Database Design & Data Modeling

Column	Data Type	Constraint	Description
full_name	VARCHAR(100)	NOT NULL	Display name
avatar_url	TEXT	NULLABLE	Profile picture URL
is_active	BOOLEAN	DEFAULT TRUE	Account status flag
created_at	TIMESTAMP	DEFAULT NOW()	Registration timestamp
updated_at	TIMESTAMP	DEFAULT NOW()	Last update timestamp

Table: profiles

Column	Data Type	Constraint	Description
- profile_id UUID	UUID	PRIMARY KEY	Unique row identifier
user_id	UUID	FK - users.user_id ON DELETE CASCADE	Owner reference
cv_url	TEXT	NULLABLE	S3 path to uploaded CV PDF
github_url	VARCHAR(255)	NULLABLE	GitHub profile URL
linkedin_pdf	TEXT	NULLABLE	S3 path to LinkedIn PDF export
experience_years	INT	DEFAULT 0	Inferred or manually set years
bio	TEXT	NULLABLE	Short personal bio
updated_at	TIMESTAMP	DEFAULT NOW()	Last profile edit

Table: skills

Column	Data Type	Constraint	Description
- skill_id UUID	UUID	PRIMARY KEY	Unique row identifier
name	VARCHAR(100)	UNIQUE NOT NULL	e.g. 'React', 'SQL'
category	VARCHAR(50)	NOT NULL	e.g. 'Frontend', 'DevOps'
level	ENUM	beginner intermediate advanced	Recommended starting level
description	TEXT	NULLABLE	Brief skill overview
is_active	BOOLEAN	DEFAULT TRUE	Soft-delete flag

Table: user_skills

Column	Data Type	Constraint	Description
- id UUID	UUID	PRIMARY KEY	Unique row identifier
user_id	UUID	FK - users.user_id	Skill owner
skill_id	UUID	FK - skills.skill_id	Skill reference
proficiency	ENUM	beginner intermediate advanced	Self-assessed or exam-verified

Column	Data Type	Constraint	Description
source	ENUM	cv_parsed manual exam_verified	How skill was added
added_at	TIMESTAMP	DEFAULT NOW()	When skill was added

Table: career_paths

Column	Data Type	Constraint	Description
- career_id UUID	UUID	PRIMARY KEY	Unique row identifier
name	VARCHAR(100)	UNIQUE NOT NULL	e.g. 'Backend Developer'
slug	VARCHAR(100)	UNIQUE NOT NULL	URL-friendly name
description	TEXT	NOT NULL	What this career entails
category	VARCHAR(50)	NOT NULL	e.g. 'Engineering', 'Data'
is_active	BOOLEAN	DEFAULT TRUE	Published flag

Table: roadmaps

Column	Data Type	Constraint	Description
- roadmap_id UUID	UUID	PRIMARY KEY	Unique row identifier
user_id	UUID	FK - users.user_id	Owner
career_id	UUID	FK - career_paths.career_id	Target career
title	VARCHAR(200)	NOT NULL	Auto-generated title
status	ENUM	active completed archived	Roadmap lifecycle state
total_nodes	INT	NOT NULL	Total skill count
completed_nodes	INT	DEFAULT 0	Progress counter
created_at	TIMESTAMP	DEFAULT NOW()	Generation timestamp

Table: roadmap_nodes

Column	Data Type	Constraint	Description
- node_id UUID	UUID	PRIMARY KEY	Unique row identifier
roadmap_id	UUID	FK - roadmaps.roadmap_id CASCADE	Parent roadmap
skill_id	UUID	FK - skills.skill_id	Skill reference
order_index	INT	NOT NULL	Ordering within roadmap

Column	Data Type	Constraint	Description
status	ENUM	locked unlocked in_progress in_review completed failed	Node state
unlocked_at	TIMESTAMP	NULLABLE	When prerequisites were met
completed_at	TIMESTAMP	NULLABLE	When skill was mastered
xp_awarded	INT	DEFAULT 0	XP given on completion

Table: courses

Column	Data Type	Constraint	Description
- course_id UUID	UUID	PRIMARY KEY	Unique row identifier
skill_id	UUID	FK - skills.skill_id	Related skill
title	VARCHAR(255)	NOT NULL	Course title
provider	VARCHAR(100)	NOT NULL	e.g. 'Udemy', 'Coursera'
url	TEXT	NOT NULL	Direct course link
duration_hrs	FLOAT	NULLABLE	Estimated hours
level	ENUM	beginner intermediate advanced	Audience level
rating	FLOAT	CHECK 0-5	Aggregated rating
source	ENUM	curated ai_recommended user_submitted	How added

Table: skill_exams

Column	Data Type	Constraint	Description
- exam_id UUID	UUID	PRIMARY KEY	Unique row identifier
skill_id	UUID	FK - skills.skill_id	Exam topic
user_id	UUID	FK - users.user_id	Test taker
questions	JSONB	NOT NULL	Generated questions array
answers	JSONB	NULLABLE	User submitted answers
score	INT	NULLABLE	Percentage score (0-100)
passed	BOOLEAN	NULLABLE	score - pass_mark
gap_report	JSONB	NULLABLE	AI gap analysis result
taken_at	TIMESTAMP	DEFAULT NOW()	Submission timestamp

Table: gamification

Column	Data Type	Constraint	Description
- gamif_id UUID	UUID	PRIMARY KEY	Unique row identifier
user_id	UUID	FK - users.user_id UNIQUE	One per user
xp_points	INT	DEFAULT 0	Accumulated XP
level	INT	DEFAULT 1	Current level (1-50)
badges	JSONB	DEFAULT '[]'	Array of earned badge IDs
streak_days	INT	DEFAULT 0	Consecutive study days
last_activity	TIMESTAMP	NULLABLE	Last learning action

5.3 Normalization & Design Decisions

Form / Topic	Notes
1NF	All tables have a single primary key; all column values are atomic. JSONB columns (questions, answers, gap_report) store structured arrays but are treated as opaque blobs - not queried column-by-column, maintaining 1NF at the relational level.
2NF	No partial dependencies exist. All non-key attributes depend on the full primary key. Junction tables (user_skills, roadmap_nodes) use composite logic via separate UUID PKs with FK constraints ensuring full dependency.
3NF	No transitive dependencies. career_paths details are not duplicated in roadmaps - only career_id FK is stored. Skill details are not repeated in user_skills - only skill_id FK is used.
Indexes	B-tree indexes on: users.email (UNIQUE), user_skills(user_id, skill_id), roadmap_nodes(roadmap_id, order_index), skill_exams(user_id, skill_id). Full-text index on skills.name for search.
Normalization Trade-off	gamification.badges uses JSONB (denormalized) rather than a badges join table. This trades strict normalization for query performance - badge reads are O(1) and never queried relationally.

6. Data Flow Diagrams (DFD)

6.1 Level 0 - Context Diagram

Shows CareerGPS as a single system process with all external entities and top-level data flows.

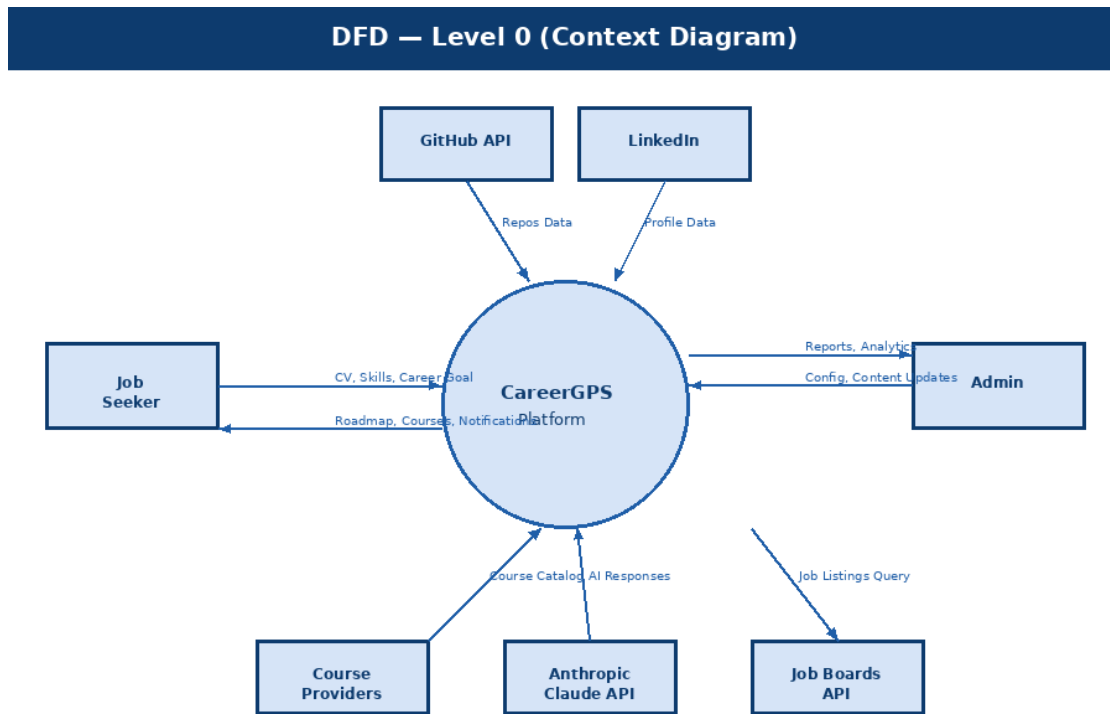


Figure 4 - DFD Level 0: Context Diagram

6.2 Level 1 - Internal System Processes

Decomposes the system into 9 internal processes showing how data flows between profile management, roadmap generation, assessment, gamification, notifications, and external APIs.

DFD — Level 1 (Internal System Processes)

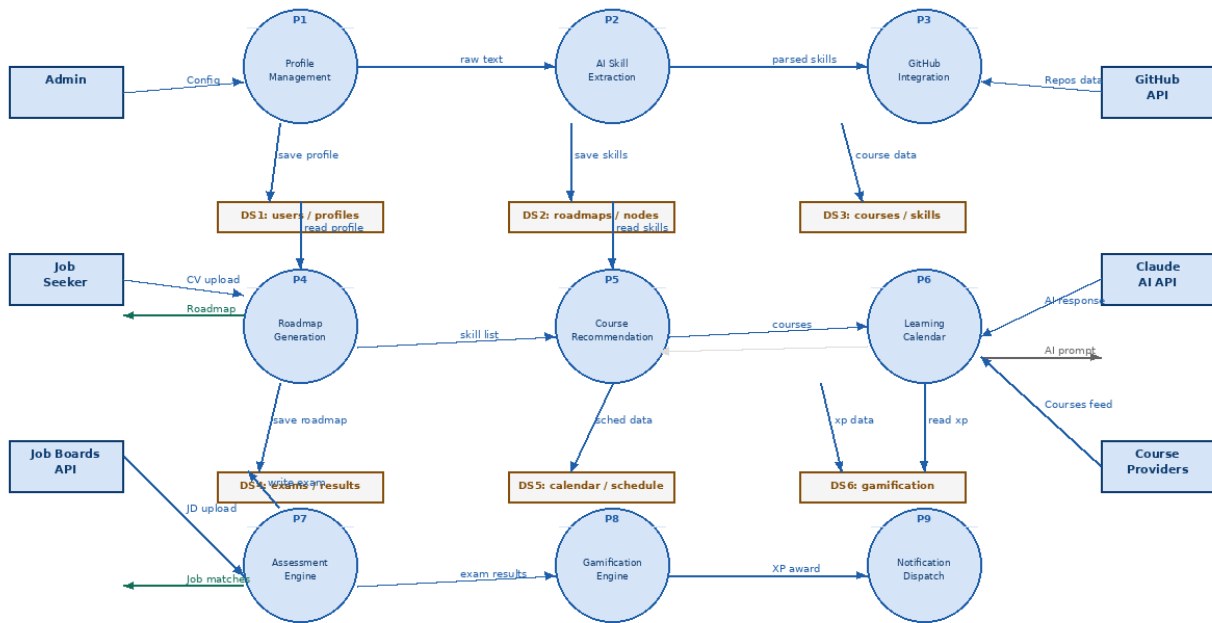


Figure 5 - DFD Level 1: Expanded Internal Processes (P1-P9 with 6 Data Stores)

- DS1: users/profiles | DS2: roadmaps/nodes | DS3: courses/skills | DS4: exams/results | DS5: calendar/schedule | DS6: gamification

7. Sequence Diagrams

7.1 Roadmap Generation Flow

End-to-end interaction from CV upload through AI processing to roadmap display. Covers 15 message exchanges across 8 system components.

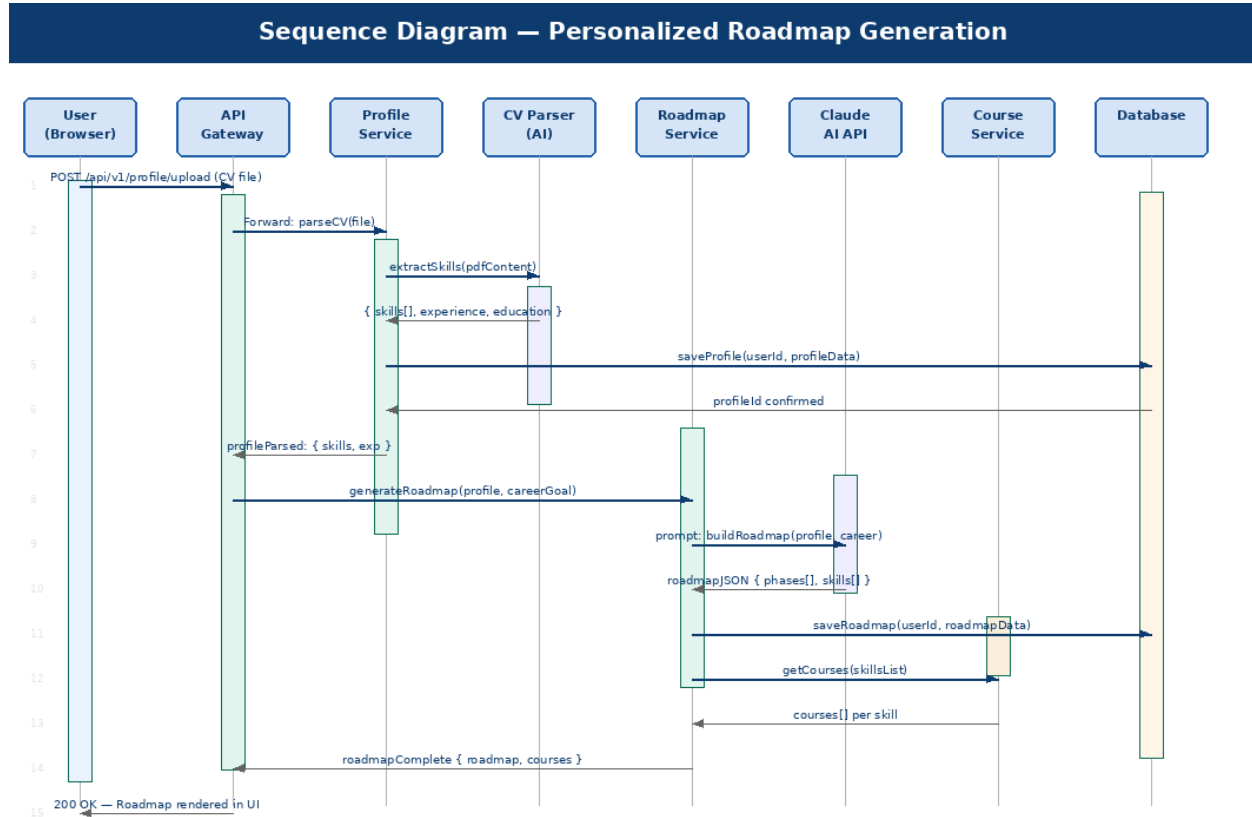


Figure 6 - Sequence Diagram: CV Upload - AI Roadmap Generation - Display

7.2 Skill Exam & Gap Analysis Flow

Covers exam session creation, question generation via Claude AI, answer submission and evaluation, gap analysis computation, and result persistence.

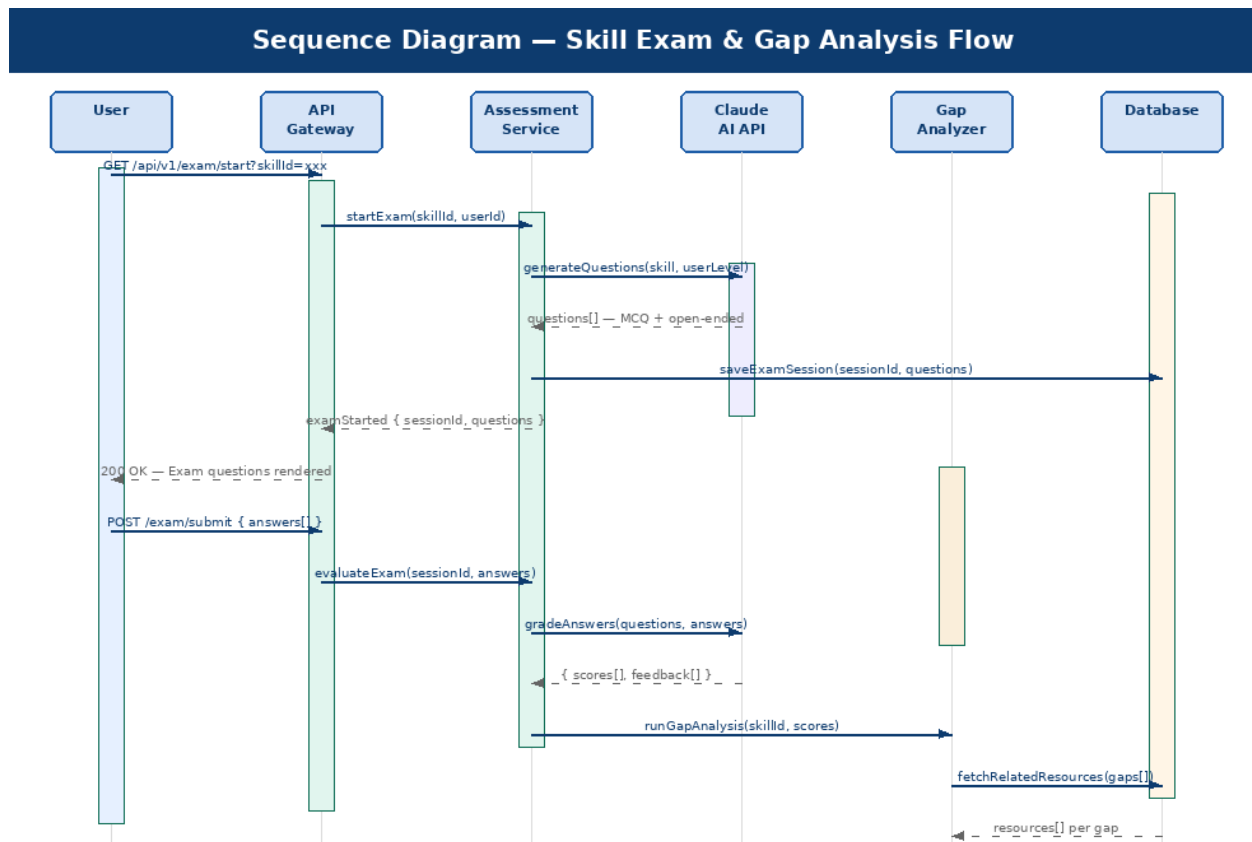


Figure 7 - Sequence Diagram: Skill Exam Start - Evaluate - Gap Analysis Report

7.3 ATS CV Builder Flow

Covers job description ingestion, keyword extraction, CV content generation, ATS scoring, file storage, and download link delivery.

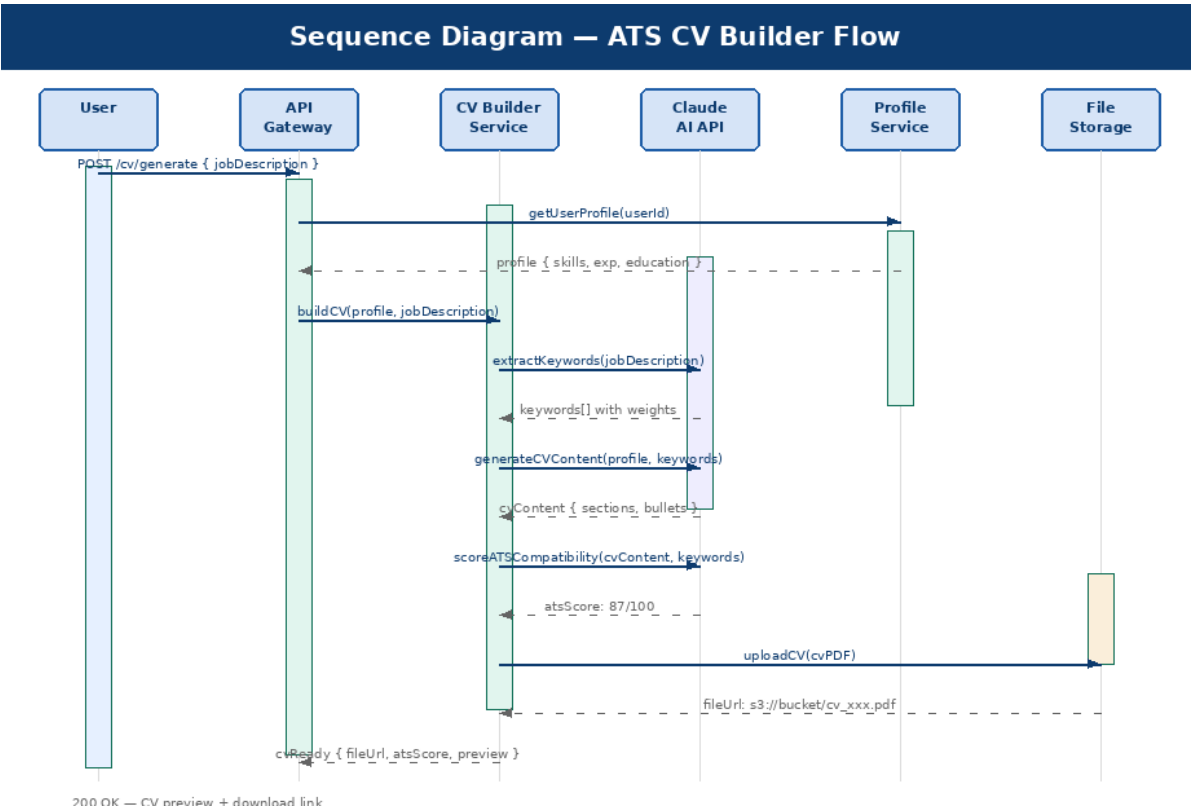


Figure 8 - Sequence Diagram: Job Description Upload - ATS CV Generation - Download

8. Activity Diagrams

8.1 User Onboarding & Roadmap Flow

Models the complete user journey from registration through CV upload, skill extraction, career selection, roadmap generation, and active learning with gamification rewards.

Activity Diagram — User Onboarding & Roadmap

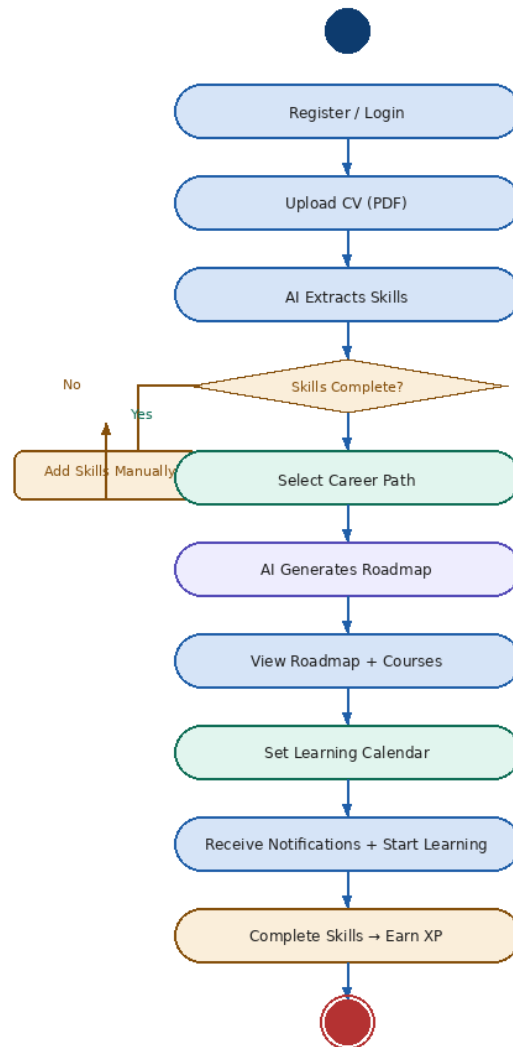


Figure 9 - Activity Diagram: User Onboarding & Roadmap Lifecycle

8.2 Interview Simulation Flow

Models the interview session flow from mode selection through AI question generation, per-answer evaluation, skip handling, and final feedback report generation.

Activity Diagram — Interview Simulation Flow

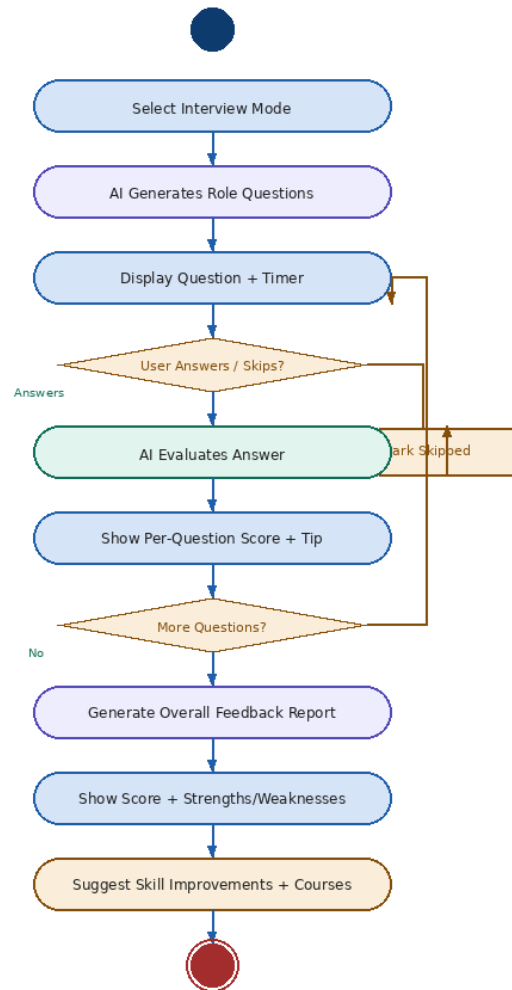


Figure 10 - Activity Diagram: Interview Simulation Session

9. State Diagrams

9.1 Skill Node Lifecycle

Each skill node in a roadmap transitions through 6 defined states. State transitions are triggered by user actions (starting, submitting) and system events (exam grading, prerequisite completion).

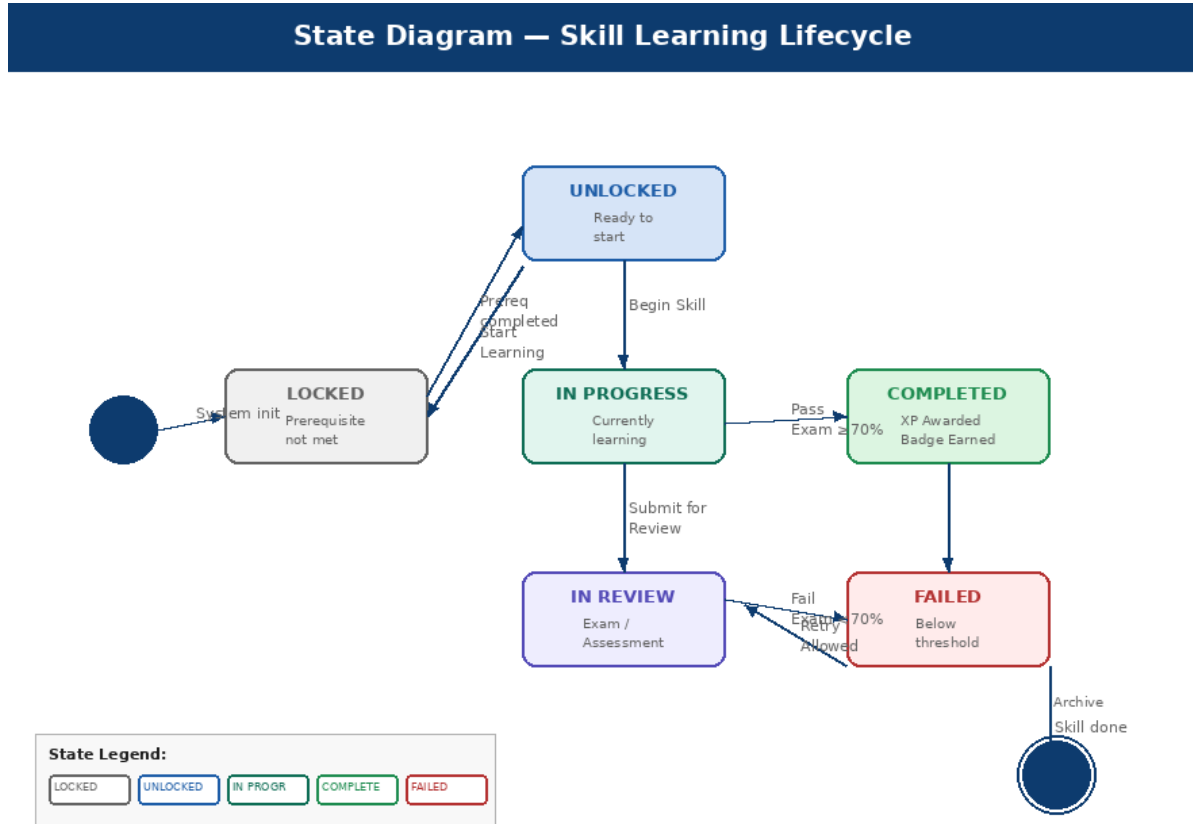


Figure 11 - State Diagram: Roadmap Skill Node (LOCKED - COMPLETED / FAILED)

State	Description	Transition Trigger
LOCKED	Skill inaccessible; prerequisites not met	Previous node marked COMPLETED
UNLOCKED	Skill accessible and ready to start	Prerequisites satisfied
IN PROGRESS	Actively learning the skill	User clicks 'Start Skill'
IN REVIEW	Exam submitted for grading	User submits exam answers
COMPLETED	Skill mastered; XP + badge awarded	Exam score - 70%
FAILED	Exam score below threshold	Exam score < 70%; retry after 24h

9.2 User Account Lifecycle

Models user account states from anonymous visitor through registration, onboarding, active usage, premium upgrade, suspension, and churn with re-engagement paths.

State Diagram — User Account Lifecycle

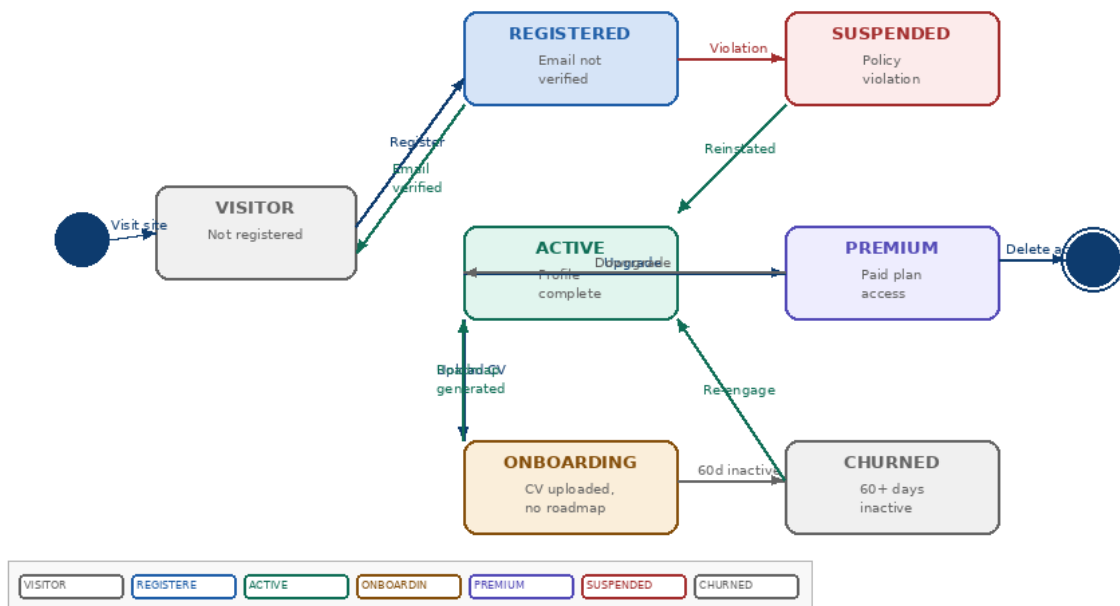


Figure 12 - State Diagram: User Account Lifecycle

10. Class Diagrams

10.1 Core Domain Model

Defines the key domain entities - User, Profile, Roadmap, Skill, Course, Exam, and their relationships. This is the blueprint for the database schema and API data transfer objects.

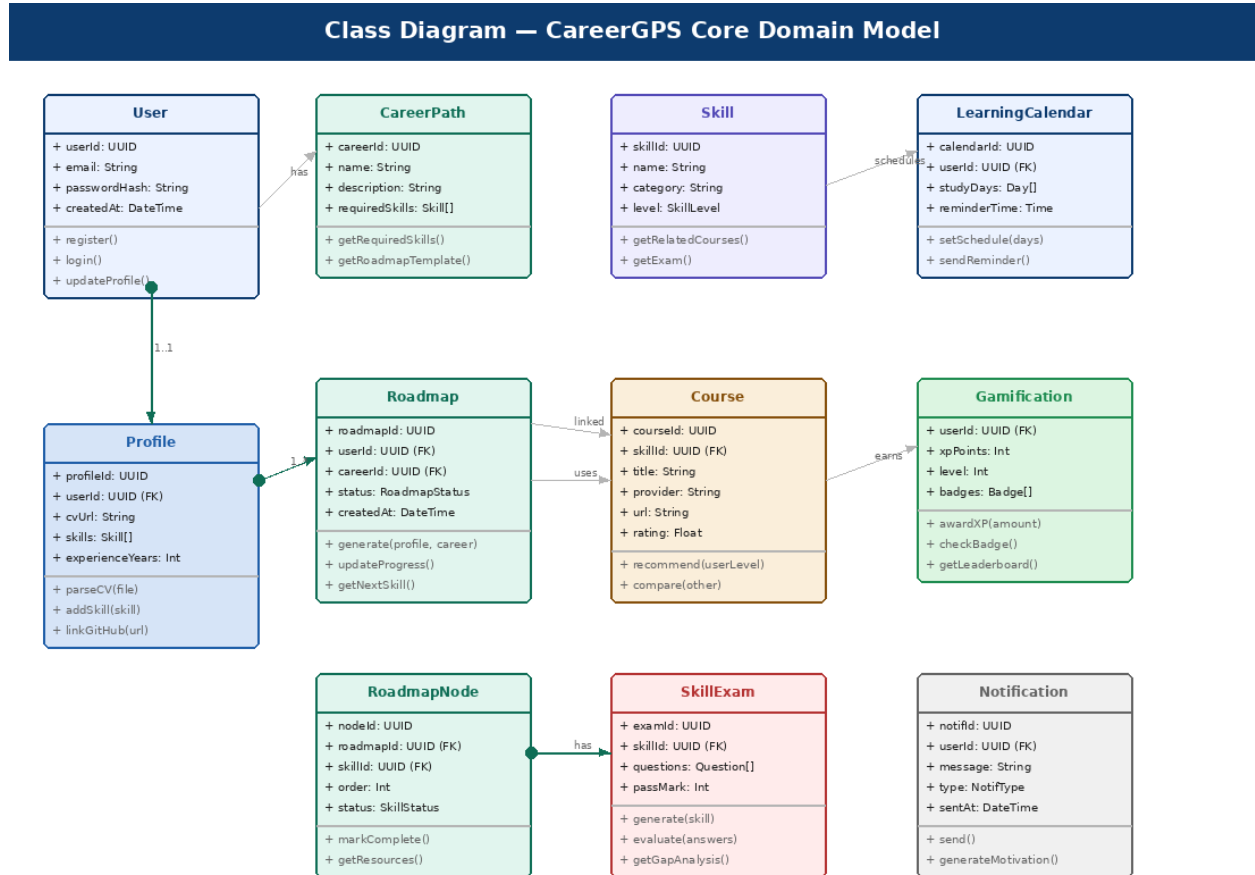


Figure 13 - Class Diagram: Core Domain Model & Entity Relationships

10.2 Key Class Relationships

Class A	Class B	Relationship	Description
User	Profile	1:1 Composition	Each user has exactly one profile; deleted with user
User	Roadmap	1:N Composition	A user can have multiple roadmaps for different careers
Roadmap	RoadmapNode	1:N Composition	Each roadmap has an ordered list of skill nodes
RoadmapNode	Skill	N:1 Association	Each node references one global skill

Class A	Class B	Relationship	Description
Skill	Course	1:N Association	Each skill has multiple curated course options
Skill	SkillExam	1:N Association	Multiple exam attempts per skill, one per user session
User	Gamification	1:1 Association	One gamification record per user - XP, level, badges
User	CareerPath	N:M Association	Users can explore multiple careers; careers have many users

10.3 Agentic AI Services Layer

The AI layer uses an -interface- pattern where all 7 AI agents implement a common AIService interface, and each agent depends on a shared ClaudeAPIClient wrapper for Claude API calls. This enforces consistent error handling, token management, and retry logic across all AI features.

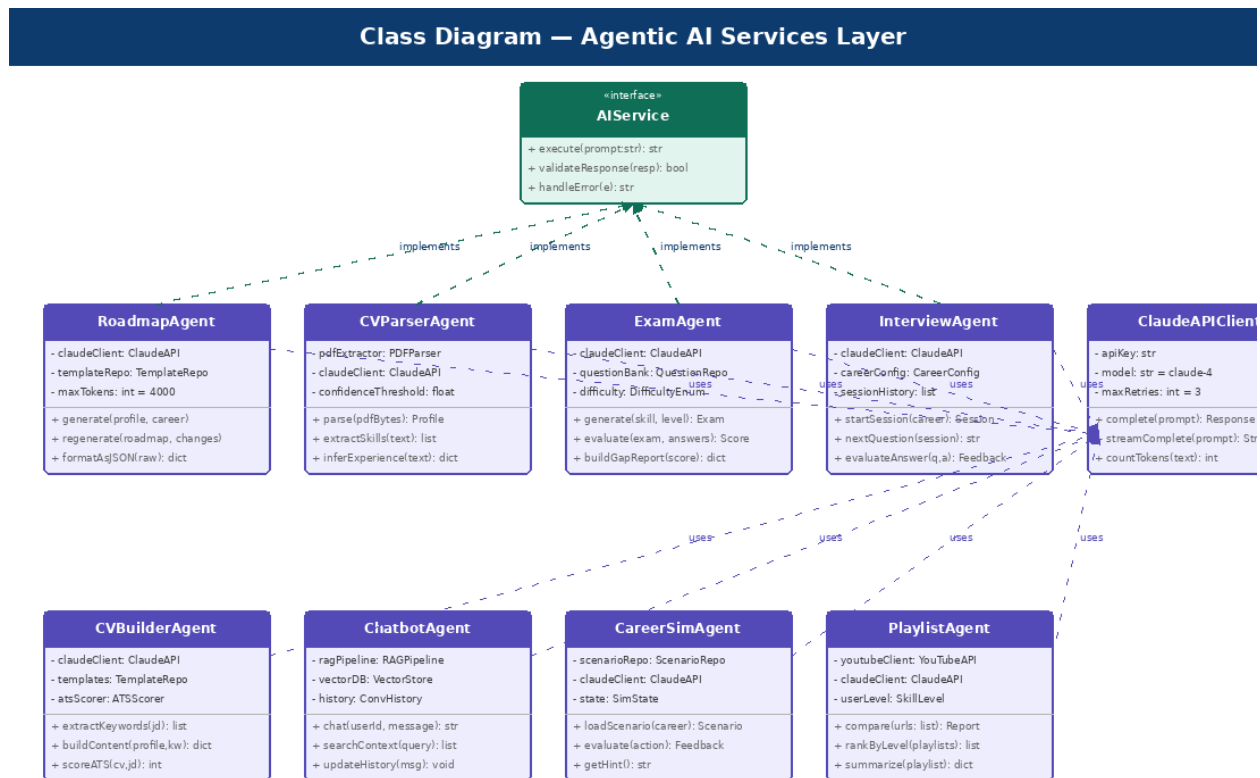


Figure 14 - Class Diagram: Agentic AI Services Layer (7 AI Agents + Interface)

AI Agent	Responsibility
RoadmapAgent	Generates personalized career roadmaps from user profile + target career using Claude
CVParserAgent	Extracts structured skills, experience, and education from raw CV PDF text

AI Agent	Responsibility
ExamAgent	Generates skill exams (MCQ + open), evaluates answers, produces gap analysis reports
InterviewAgent	Conducts mock interviews with role-specific questions and per-answer AI feedback
CVBuilderAgent	Extracts JD keywords, builds ATS-optimized CV content, scores ATS compatibility
ChatbotAgent	RAG-based conversational assistant; retrieves context from vector DB for accurate answers
CareerSimAgent	Manages guided hands-on career simulation scenarios with dynamic task evaluation
PlaylistAgent	Compares YouTube playlists by content, level match, and quality for skill learning

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